Propeller De-icing System (Mod S.O.O. 6202) – General Data

1. General

Propeller de-icing is effected by providing a cycled electrical power supply to inner and outer heating elements, contained in de-icer boots, installed on the leading edge of each propeller blade.

2. Description

The propeller de-icing system comprises the boots and slip ring assemblies which are a part of the two propeller rotating assemblies (refer to 61-10-11, Propeller De-icing (S.O.O. 6202) – General Data), a brush block assembly and oil pressure switch installed on each engine, two relays, a timer, and a control switch. Electrical power to the system is supplied from the 28 Volt DC bus, through a 20-ampere circuit breaker, identified PROPELLER DEICE, located on the circuit breaker panel. A brush module assembly replaces the brush block assembly (Mod 6/1674). Refer to Figure 1 for details.

A. Brush Module Assembly – (Mod 6/1674)

A brush module assembly is installed on a bracket secured to the front of each engine reduction gearbox. Each assembly houses three brushes which make contact with three slip rings on the rotating slip ring assembly (refer to 61-10-11, Propeller De-icing (S.O.O. 6202) – General Data). Each brush module assembly consists of a bracket and a brush retainer assembly comprised of three brush modules wired to a connector. Each brush module consists of a plastic housing with an integral brush and spring. The modules each have a wire internally connected to the brush with the three wires terminating in a single connector for connection to the propeller de-icing system.

B. Oil Pressure Switches

An oil pressure switch, which is also connected into the low oil pressure caution warning circuit (refer to 79-36-00, Low Oil Pressure Indicating System – General Data), is mounted on each engine accessory gearcase. When an engine is running, the oil pressure switch contacts are closed to provide a ground for the relevant propeller de-icing relay operating coil circuit, to allow the relay to energize when the system is selected on.

C. Relays

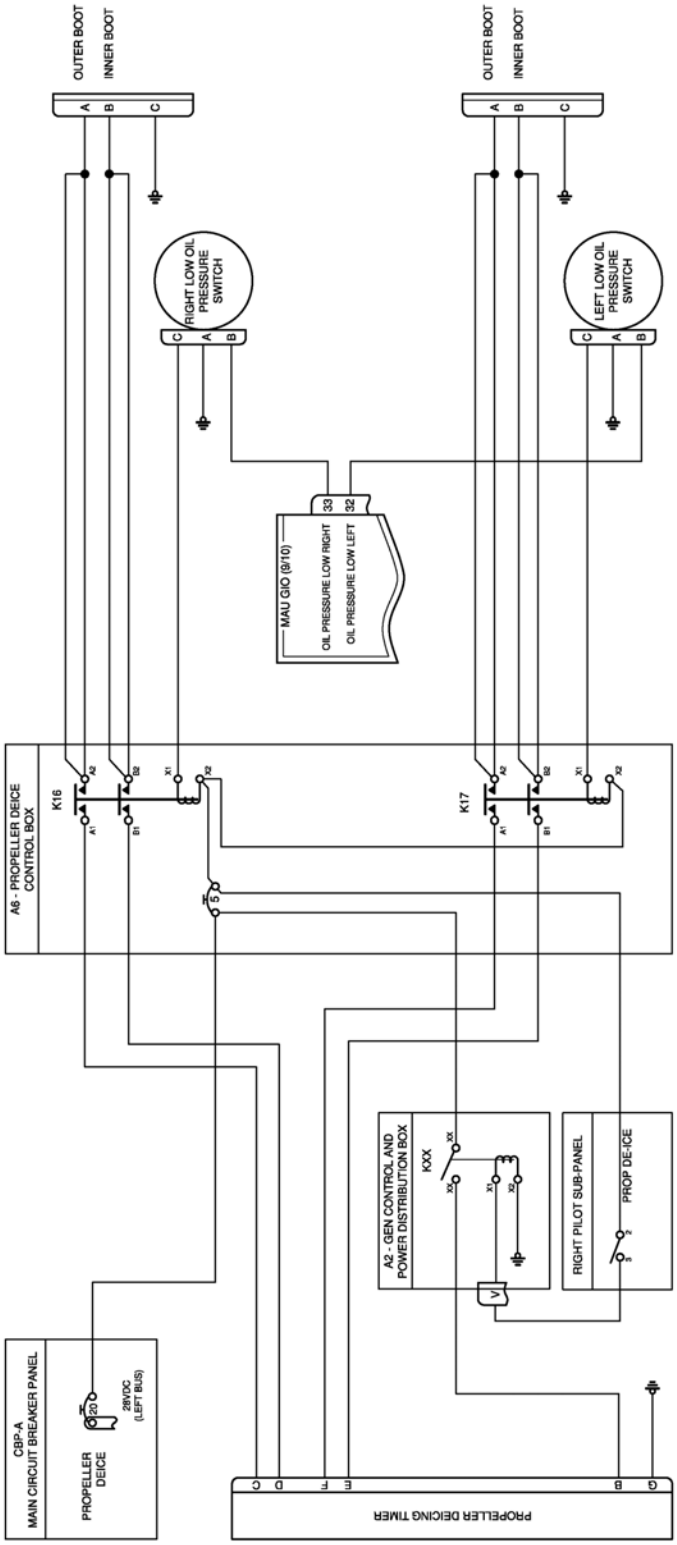
Two propeller de-icing relays, each controlling a single propeller system, are installed in the propeller de-icing control box located in the cabin roof. Each relay, when energized, completes two circuits, one from the timer to the outer heater elements in the blade boots of the associated propeller, and the other to the inner elements of the boots of the same propeller. A manual reset 5-ampere circuit breaker, connected in series with both relay control coils, is installed on the de-icing control box to protect the relay coils.

D. Timer

The timer, mounted in the cabin roof, is an electronic device which provides four sequenced power outputs, two to each propeller system.

E. Control Switch

A control switch identified PROP DE-ICE is located on the right pilot sub-panel. The switch completes the circuit to provide electrical power to the timer and each propeller de-icing relay control coil.



Propeller De-icing System – Electrical Schematic
Figure 1

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3. Operation

With both engines running, the PROPELLER DEICE circuit breaker engaged, and the PROP DE-ICE control switch selected ON, electrical power is provided to the timer, and the relays are energized resulting in a sequenced power output being provided to the inner and outer elements of both propellers in turn. In the event of a propeller being feathered, or an engine shutting down, the oil pressure switch contacts open (due to low or loss of oil pressure), and the appropriate propeller de-icing relay de-energizes to open the circuits to the heating elements of the affected engine's propeller.

4. Adjustment/Test

A. Operational Test Propeller De-icing System (with engines running)

Note It is essential that the propeller de-icer brushes are properly run in on their slip rings before carrying out the test. After installation, whether new or replacement brushes are used, allow a run-in period of at least one hour.

1) Engage PROPELLER DEICE circuit breaker.

2) Start engines, turn on one generator and turn off electrical equipment not required. Observe limitations and procedures in accordance with 71-00-00, Power Plant (Mod 6/2012) – General Data.

3) Select PROP DE-ICE switch on right pilot's sub-panel to ON.

4) Check for correct functioning of propeller heating circuits as shown by an initial increase in generator load of approximately 6 amps.

Note Observe that the current remains approximately as above for 2 minutes and 30 seconds to verify that the system cycles through all four heating circuits.

5) Switch PROP DE-ICE switch to OFF.

6) Shutdown engines in accordance with 71-00-00, Power Plant (Mod 6/2012) – General Data.

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